Freight Cargo Inventory and Shipment Management System

Project Progress Report 1 - EERD

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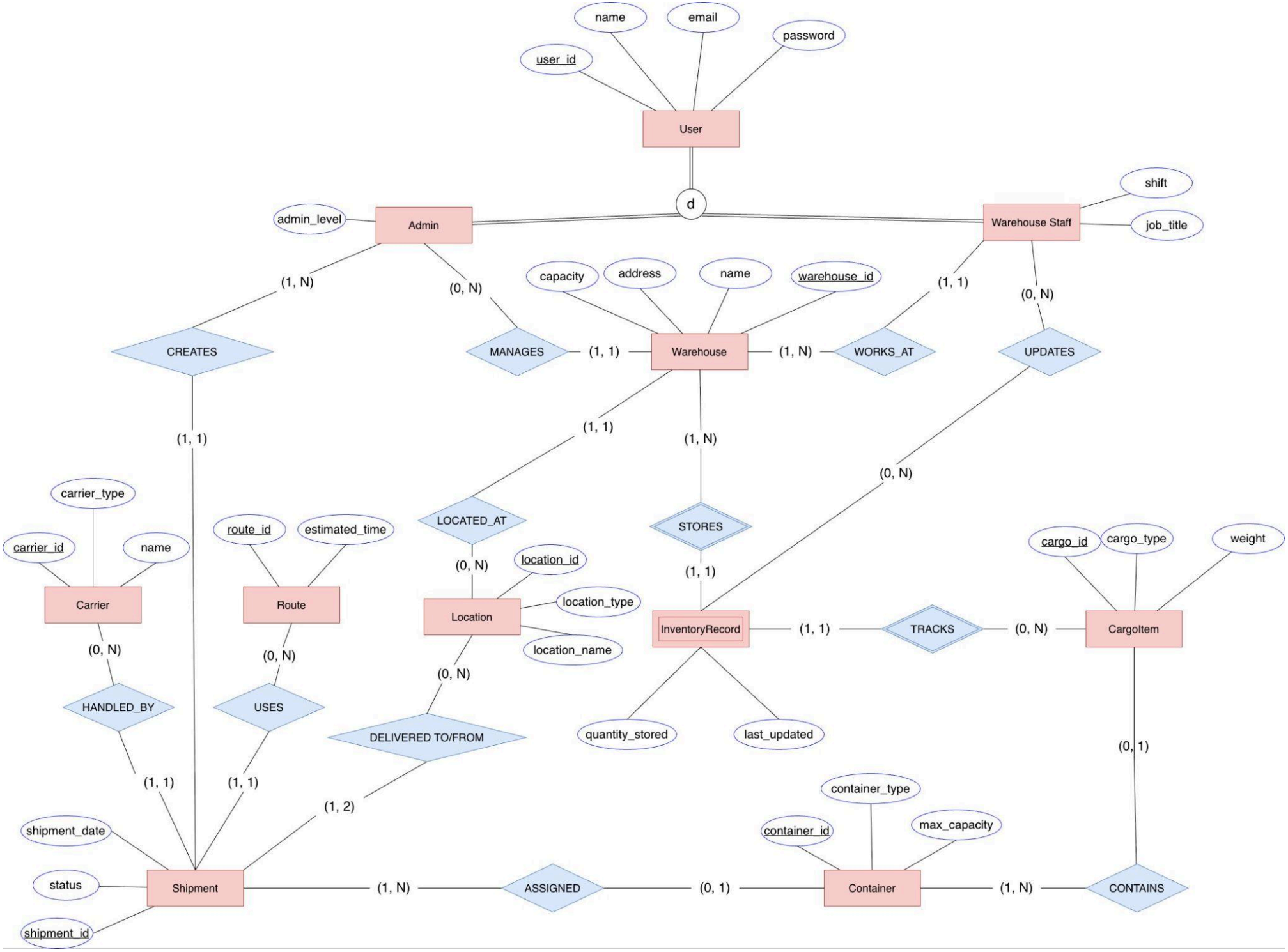
CPSC471 W26

Group G-4

Assumptions:

1. Every user is either an Admin or a Warehouse Staff member, never both and never neither
2. Email is unique across all users
3. Each warehouse is managed by exactly one Admin
4. Each Warehouse Staff member works at exactly one warehouse
5. Each shipment is handled by exactly one carrier and created by exactly one Admin
6. Each shipment uses exactly one route
7. A route has exactly one origin and one destination location
8. A container can be assigned to at most one shipment at a time
9. A cargo item can be stored in at most one container at a time
10. InventoryRecord is a weak entity, identified by its relationship to Warehouse
11. InventoryRecord is uniquely identified by (warehouse_id, cargo_id).
12. Each InventoryRecord tracks exactly one CargoItem
13. Warehouse Staff may update zero or more inventory records
14. Carriers are pre-existing external entities registered in the system by Admins
15. The system only tracks the current container-to-shipment assignment, not historical assignments
16. A cargo item's warehouse location is determined through its InventoryRecord, and its in-transit location is determined through its container's shipment


name
email
password
user_id
User
d
admin_level
Admin
Warehouse Staff
shift
job_title
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(0, N)
capacity
address
name
warehouse_id
(1, 1)
(0, N)
CREATES
MANAGES
(1, 1)
Warehouse
(1, N)
WORKS_AT
UPDATES
(1, 1)
(1, 1)
(1, N)
(0, N)
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LOCATED_AT
STORES
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cargo_type
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Route
Location
location_type
InventoryRecord
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(0, N)
HANDLED_BY
USES
DELIVERED TO/FROM
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(1, 1)
(1, 2)
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shipment_date
container_id
max_capacity
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Shipment
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ASSIGNED
(0, 1)
Container
(1, N)
CONTAINS
shipment_id